

Lab #4 - Pointers & Functions: Digital Time Display

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In this lab, you will complete a partially provided C program that prints a digital time display using 3×3 ASCII digits.

Example Digits (0-9):

_	_	_	_	_	_	_	_	_	_
_	_	_	_	_	_	_	_	_	_

You will implement several functions that use **pointers** to return multiple values, and you will validate user input in `main`.

Part 1: Printing Digital Time (3×3 ASCII Digits)

(40 points)

In this part, you will complete the functions responsible for printing the time in a digital 3×3 ASCII format.

(a) `char digitCell(int digit, int row, int col);`

Complete the missing cases for the digits **4**, **6**, and **9** of the function returning the character to be printed at position (`row`, `col`) for a given digit.

Inputs: `digit` (0–9), `row` (0–2), `col` (0–2).

Output: Returns ' ', '|', or '_' for the given cell.

(b) `void split2Digits(int value, int *tens, int *ones);`

Split the given value (0–99) into its tens and ones digits using pointer parameters.

Output: Stores the tens digit in `*tens` and the ones digit in `*ones`.

(c) `void printTime3x3(int h, int m, int s);`

Inside this function, obtain the six digits of the time (HHMMSS) by calling `split2Digits` exactly three times: (`h` → `ht`, `ho`), (`m` → `mt`, `mo`), (`s` → `st`, `so`).

Inputs: `h`, `m`, `s` (time values).

Output: Prints the digital time in a 3×3 ASCII format.

Part 2: Converting Seconds to Time (HH:MM:SS)

(20 points)

In this part, you will complete the function responsible for converting the total number of seconds into hours, minutes, and seconds.

(a) `void convertTime(int totalSec, int *h, int *m, int *s);`

Convert `totalSec` into hours, minutes, and seconds using pointer parameters.

Input: `totalSec` (0–86399).

Output: Stores the results in `*h`, `*m`, and `*s`.

Part 3: Completing main

(40 points)

In this part, you will complete the `main` function. Your program should (i) read input correctly, (ii) call `convertTime`, and (iii) call `printTime3x3`.

(a) Read a valid input:

Ask the user to enter `totalSeconds`. If the value is not in the range **0–86399**, print an error message and ask again. Continue until a valid value is entered.

(b) Call `convertTime`:

Declare `int h, m, s;` and call the conversion function using pointer arguments.

(c) Call `printTime3x3`:

Finally, call the printing function to display the digital time.

Expected program executions:

```
Enter total seconds (0-86399): 3661
```

```
  _  _  _  _  _
 | | | | | | |
 | _| | _| | _|
```

```
Enter total seconds (0-86399): -5
```

```
Invalid input! Please enter a value between 0 and 86399.
```

```
Enter total seconds (0-86399): 90000
```

```
Invalid input! Please enter a value between 0 and 86399.
```

```
Enter total seconds (0-86399): 86399
```

```
  _  _  _  _  _  _
 | _| | _| | _| | _|
 | _| | _| | _| | _|
```